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MILITARY SPECIFICATION

PUMPS, RECIPROCATING, POWER-DRIVEN: DIAPHRAGM, GASOLINE-ENGINE-DRIVEN, WHEEL-MOUNTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers gasoline-engine-driven wheel-mounted diaphragm pumps (hereinafter called "pump units").

1.2 Classification. Pump units shall be the following classes and sizes, as specified (see 6.2):

Class 1 - Driven by a commercial engine.
Class 2 - Driven by a Government-furnished engine.
Size A - 50 gallons per minute (gpm) capacity.
Size B - 100 gpm capacity.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060 by using the self-addressed Stand-Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4320

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-601	- Aluminum Alloy Sand Castings.
QQ-S-741	- Steel, Carbon: Structural Shapes, Plates; and Bars.
TT-V-119	- Varnish, Spar, Phenolic-Resin.
UU-T-81	- Tags, Shipping and Stock.
VV-G-1690	- Gasoline, Automotive, Leaded or Unleaded.
WW-C-624	- Coupling Assembly, Hose (Garden, Water, and Water Suction).
WW-N-351	- Nipples, Pipe, Threaded.
ZZ-H-561	- Hose, Rubber, and Hose Assemblies, Rubber, Smooth Bore, Water Suction and Discharge.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-60	- Tape: Packaging, Waterproof.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-R-196	- Repair Parts, Accessories, and Kits, Mechanical, Packaging of.
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-P-775	- Packaging of Hose, Hose Assemblies; Rubber, Plastic, Fabric, or Metal (Including Tubing): and Fittings, Nozzles, and Strainers.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.

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MIL-F-3541	- Fittings, Lubrication.
MIL-G-3859	- Grease Guns, Hand-Operated, Lever, Push and Screw Type.
MIL-E-10062	- Engines: Preparation for Shipment and Storage of.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-V-13811	- Varnish, Waterproofing, Electrical, Ignition.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-P-23377	- Primer Coatings: Epoxy-Polyamide, Chemical and Solvent Resistant.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing: With Appropriate Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MIL-STD-1227	- Engine, Gasoline: Air-Cooled, 3 BHP, 4-Cycle, Military Design, Model 2A016, Installation Procedures.
MIL-STD-1400	- Engine, Gasoline or Diesel, Methods of Test.
MIL-STD-1474	- Noise Limits for Army Materiel.

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MS35844	- Plug, Machine Thread, Magnetic: Drain.
MS49006	- Plugs, Pipe, Magnetic-Headless; Aluminum, Iron or Steel.
MS51086	- Strainer, Sediment; Automotive Fuel (10 GPH).

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASME - Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
ANSI/AWS D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 48 - Grey Iron Castings.
ASTM A 242 - High-Strength Low-Alloy Structural Steel.
ASTM D 2000 - Rubber Products in Automotive Applications.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

HYDRAULIC INSTITUTE (HI)

Rating Standards and Test Code Standards.

(Application for copies should be addressed to the Hydraulic Institute, 712 Lakewood Center North, Cleveland, OH 44107.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15086.)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The pump units shall consist of a diaphragm pump, a speed reduction system, a gasoline engine, and the necessary accessories and connections. The pump units shall be wheel mounted. The pump units shall, as a minimum, be in accordance with the requirements of this specification and shall be the contractor's standard commercial product (see 6.4) with any added features needed to comply with the requirements. Unless otherwise specified herein, the components shall also be the contractor's standard commercial product when they meet the requirements of this specification. Additional or better features are not specifically prohibited by this specification.

3.2 First article. When specified, (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. The material shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the pump units to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product (see 6.13).

3.3.1 Material deterioration and control. The pump units shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.1.1 Dissimilar Metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a coat of epoxy primer conforming to MIL-P-23377, or corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

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3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.4 Threads. All screw threads shall be in accordance with FED-STD-H28.

3.5 Safety. All exposed parts that are subject to high operating temperatures or that are energized electrically shall be insulated, fully enclosed, or guarded. All moving parts that are of such nature or so located as to be a hazard to operating or maintenance personnel shall be fully enclosed or guarded. Protective devices shall not impair the operating functions.

3.6 Noise limits. The noise of the pump units shall not exceed the limits specified in MIL-STD-1474, category D, for steady-state noise when tested in accordance with 4.5.2.4 (aural non-detectability and impulse noise do not apply). The precautions of MIL-STD-1474 (see 6.5) shall be provided if the sound level exceeds 85 dB(A) when tested in accordance with 4.5.2.4. In addition the distance from the pump unit at which a limit of 85 dB(A) is met shall be shown on hazard signs mounted on all four sides of each production unit stating: "CAUTION, HEARING PROTECTION REQUIRED WITHIN _____ FEET WHEN EQUIPMENT IS OPERATING". The caution shall be legible from the distance shown on the sign. Hearing protectors shall not be supplied with the pump units. Plates shall be in accordance with MIL-P-514, type III, composition C of type I, grade A, class 1 material."

3.7 Reliability.

3.7.1 Mean-time-between-failure (MTBF). The specified MTBF of the pump unit, exclusive of the diaphragm, shall be 140 hours when tested as specified in 4.5.2.5.1.

3.7.2 Diaphragm. The specified MTBF of the diaphragm shall be 100 hours when tested as specified in 4.5.2.5.2.

3.8 Maintainability.

3.8.1 Maintenance support. Assemblages or support elements such as technical manuals, repair parts, special tools, or lubrication charts, as specified (see 6.9 and 6.10), shall be adequate to perform the intended function of assisting or conducting maintenance operations when tested as specified in 4.5.2.6.

3.8.2 Maintenance ratio. The pump unit shall have a maintenance ratio of not more than 0.05. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hours for repair of replaced components and scheduled before-and-after operation checks are excluded. A maintenance schedule shall be furnished prior to the start of any testing.

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3.9 Pump. The pump shall be diaphragm lift-and-force type, shall be self-priming, and shall pump water containing solids. The pump shall be equipped with a suction passageway, suction valve with replaceable valve seat, suction-valve chamber, diaphragm chamber, flexible diaphragm, discharge valve with replaceable valve seat, discharge-valve chamber, and discharge passageway. Valves shall be integral flapper or ball type. Means shall be provided (without uncoupling the suction or discharge hoses) for removing obstructions that might prevent the valves from seating properly. The suction and discharge ports shall be as specified for each size of pump.

3.9.1 Pump case. The pump case shall be either an iron casting conforming to ASTM A 48, class 20A or higher, shall be fabricated of steel having physical requirements in accordance with ASTM A 242, or an aluminum casting conforming to QQ-A-601, alloy 535.0, temper optional.

3.10 Speed-reduction system. Power from the engine shall be transmitted to the pump through a gear train, a roller-chain drive or a flexible spline drive enclosed in an oiltight case. The rotating motion shall be changed to a reciprocating motion at the diaphragm by means of an eccentric crank, cam, or walking beam. The speed-reduction system shall drive the pump plunger and diaphragm at a normal operating rate of 60 strokes per minute. Bearings used in the speed-reduction system shall be either the ball or roller type. When specified (see 6.2), power from the engine shall be transmitted to the pump through a gear train.

3.11 Performance based on standard conditions. At a speed of not more than 60 strokes per minute, the pump shall deliver the capacities specified herein for each size of pump unit. Each pump unit shall pump against a total discharge head of 35 feet with a static suction lift of 25 feet. The pump shall not leak when subjected to a hydrostatic pressure of not less than 50 pounds per square inch (psi). Standard conditions are defined as an atmospheric pressure of 29.92 inches of mercury and a water temperature of not less than 68° F.

3.12 Base frame and wheel mounting. The base frame and wheel mounting shall consist of a structural steel base or a base integral with a diaphragm-retainer casting, an axle, two wheels, a removable handlebar, and a support to hold the unit in a horizontal position when stationary. When the base frame is fabricated of structural steel, the steel shall conform to QQ-S-741. The frame shall be welded into a rigid support for the engine, the pump, and the speed-reduction system. The pump unit shall be mounted in a balanced position over the axle and shall be bolted to the base. The wheels shall be mounted on antifriction bearings and furnished with cushion tires, size 12.00 x 3 x 2 ply. When specified (see 6.2), wheels shall be furnished with cushion tires not less than size 10 x 2.75 x 2 ply.

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3.13 Size A pump unit.

3.13.1 Suction and discharge ports. The suction and discharge ports shall be internally threaded with 3-inch NPT threads. Each port shall be provided with an externally threaded 3-inch NPT to 3-inch NH pipe-to-hose adapter made of brass or zinc-plated steel.

3.13.2 Performance. At a speed of not more than 60 strokes per minute, the pump shall deliver not less than 50 gpm at a static suction lift equivalent to 10 feet and 33 gpm at a static suction lift equivalent to 20 feet.

3.14 Size B pump unit.

3.14.1 Suction and discharge ports. The suction and discharge ports shall be internally threaded with 4-inch NPT threads, and each port shall be provided with a zinc-coated wrought-iron or steel 4-inch pipe nipple conforming to WW-N-351, type I, 6 inches long with 4-inch NPT threads.

3.14.2 Performance. At a speed of not more than 60 strokes per minute, the pump shall deliver not less than 100 gpm at a static suction lift equivalent to 10 feet and 66 gpm at a static suction lift equivalent to 20 feet.

3.14.3 Hose. When specified (see 6.2), the size B pump unit shall be provided with 4 lengths of suction hose conforming to ZZ-H-561, grade B, type I. The hose shall be 4 inches in diameter and 10 feet in length and shall be equipped with couplings conforming to WW-C-624, type B, style 1, except that the threads shall be 4-inch NPT.

3.15 Engine.

3.15.1 Class 1. The gasoline engine shall be a heavy-duty 4-cycle, air-cooled commercial-industrial type capable of operating on gasoline conforming to VV-G-1690 and lubricating oil conforming to MIL-L-2104. The engine shall be equipped with a heavy-duty recoil-type manual starter and a dry-type air cleaner.

3.15.1.1 Power and speed rating. The power and speed rating of the engine shall be such that operation of the pump at any capacity and under any operating conditions specified herein will not require horsepower in excess of the continuous horsepower rating of the engine as previously published by the engine manufacturer, or 85 percent of the engine manufacturer's previously published maximum horsepower rating, at the applicable speed, whichever is less.

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3.15.1.2 Fuel system. A fuel tank shall be furnished. The fuel tank shall have a capacity of not less than 1 gallon. A fuel line shutoff valve attached directly to the tank shall be provided for gravity feed systems. A standpipe not less than 0.25 inch high from the bottom shall be incorporated in the fuel tank to prevent entry of dirt and foreign matter into the system. Fuel tanks shall be located so that spills or overflows will not run onto the engine, other hot surfaces, or electrical components. A fuel strainer conforming to MS51086 shall be furnished and shall be mounted between the fuel tank outlet and the fuel pump inlet and shall be located so that it is accessible for service. When specified (see 6.2), a diaphragm-type fuel pump shall be furnished.

3.15.1.3 Muffler. The muffler shall be positioned in such a manner that exhaust gases are directed away from the operator.

3.15.2 Class 2. The engine shall be Government-furnished, military standard, model 2A016, 3-horsepower (hp), and shall be installed in accordance with MIL-STD-1227 (see 6.7).

3.15.2.1 Fuel tank. A fuel tank shall be furnished with a fuel capacity of not less than one gallon. A fuel line shut off valve shall be attached directly to the tank to permit removal of the fuel line without draining the tank. A stand pipe not less than 0.25 inch from the bottom shall be incorporated in the fuel tank to prevent entry of dirt and foreign matter into the fuel system. The tank shall be located to provide gravity flow to the engine fuel pump and to prevent spills and overflows from running on the engine, other hot surfaces, or electrical components. The fuel strainer furnished with the Government-furnished engine shall be installed between the fuel tank and fuel pump.

3.15.2.2 Speed-reduction system. The speed-reduction system shall conform to 3.10 and be as specified herein. The speed-reduction system shall be compatible with the Government-furnished engine (see 3.15.2) and shall reduce the engine output shaft (crankshaft) speed of 3600 revolutions per minute (rpm) to 60 strokes per minute (pump speed) on the speed reducer output shaft without inducing excessive torque, torsional or flexural loadings in excess of the engine capabilities and without exceeding the requirements of MIL-STD-1227. The speed-reduction system shall be designed for minimum power loss between the engine and pump.

3.15.2.3 Power requirement. The power requirement of the pump and speed-reduction system at standard conditions (see 3.11) shall not exceed the continuous duty horsepower rating (3 hp) of the engine at 3600 rpm (see 6.8).

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3.16 Electromagnetic interference. Unless otherwise specified (see 6.2), the electromagnetic interference emission of the class 1 pump units shall conform to MIL-STD-461, class C1, group II.

3.17 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.17.1 Lubricants. The pump unit shall operate as specified herein when lubricated with military lubricants (see 6.6). When specified (see 6.2), lubricants as recommended by the manufacturer shall be furnished.

3.17.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, types I, II or III. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun conforming to MIL-G-3859 with a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts.

3.17.3 Filling, draining, and checking provisions. Enclosures such as gear-cases and transmission housings that contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with a dip-sticks, check-plugs not less than 1/2-inch pipe size, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped for filling the enclosure with a lubricant and for draining. The drain outlet shall be fitted with a magnetic drainplug, size conforming to MS35844 or MS49006. The drain shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be to a container when the pump unit is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain outlet to a container.

3.17.4 Grease lubrication. Unless otherwise specified (see 6.2), all grease lubrication, including lubrication of sealed bearings, shall be with greases as follows:

a. MIL-G-23827 - for ball, roller and needle bearings, gears, and on sliding and rolling surfaces of such equipment as mechanical control systems; for use on gears and other equipment requiring a lubricant with high load-carrying capacity over a temperature range of minus 100° F to plus 250° F.

b. MIL-G-10924 - For lubrication and surface corrosion protection of all automotive equipment operated over the temperature range of minus 65° F to plus 225° F. Not for use on machinery which comes in contact with foods or potable water.

The pump units shall be assembled, run-in (broken-in), tested, and delivered with these greases. The bearings shall be cleaned before lubrication because military greases are not always compatible with other greases. A tag shall be attached in a conspicuous place to indicate which military grease has been used.

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3.17.5 Oil lubrication. Unless otherwise specified (see 6.2), all components shall be filled to the operating level with military oils designated for use in the temperature range specified herein. A tag shall be attached in a conspicuous place to indicate the temperature range and grade of lubricant used.

3.18 Slinging provisions. Permanently affixed slinging provisions that enable the pump units to be lifted into their normal travel position shall be provided and shall conform to MIL-STD-209. Slinging provisions may also be used as tie-downs when such provisions conform to MIL-STD-209.

3.19 Tiedown provisions. Permanently affixed tiedown provisions that permit fastening the pump unit to the floor or deck of a transportation medium shall be provided and shall conform to MIL-STD-209. Tiedown provisions may also be used as slinging provisions when such provisions conform to MIL-STD-209.

3.20 Identification marking. The pump units shall be identified in accordance with MIL-STD-130. The marking shall be applied to the pump units on plates conforming to MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material. Each plate shall be attached by screws, bolts, or rivets in a location where the plate shall be both visible and legible.

3.21 Instruction plates. Each pump unit shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the pump unit. Instruction plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material. Each plate shall be attached by screws, bolts, or rivets in a location where the plate shall be both visible and legible.

3.22 Treatment and painting. The portions of the pump units and their components normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A.

3.23 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more parts possessing such functional and physical characteristics as to be equivalent in performance and durability and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts except for adjustment and without selection for fit or performance.

3.24 Workmanship.

3.24.1 Castings and forgings. All parts, components, and assemblies of the pump units that include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures that do not reduce strength or affect function.

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3.24.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.24.3 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler materials, and fabrication techniques shall be as required to enable the pump unit to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 0.1875 inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the pump units shall be at the option of the contractor.

3.24.4 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the types of welding operations to be performed and that such qualification is effective as defined by the particular code:

ANSI/AWS D1.1, Structural Welding Code, Section 5, Qualification.

ANSI/ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with ANSI/AWS or ANSI/ASME codes.

3.24.5 Bolted connections. Bolt holes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement. Bolts shall not protrude more than two full threads.

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3.24.6 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows.

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article pump units shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article pump units shall be tested as specified in column 1 of table I. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for tests shall be in accordance with MIL-STD-105, inspection level S-2.

4.4.2 Examination. Each pump unit shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

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4.4.3 Tests.

4.4.3.1 Individual. Each pump unit shall be tested as specified in column 2 of table I. Failure of any test shall be cause for rejection.

4.4.3.2 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in column 3 of table I. AQL shall be 1.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The pump units shall be examined as specified herein for the following defects:

Major

101. Dimensions not as specified.
102. Materials not as specified.
103. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable and operating environments.
104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
105. Contractor does not have documentation available for identification of material, material finishes or treatment.
106. Screw threads not as specified.
107. Safety devices or guards missing or not as specified.
108. Pump not as specified.
109. Engine not as specified.
110. Lubrication devices not as specified.
111. Electromagnetic interference suppression components and network not identical to the approved model.
112. Interchangeability not as specified.
113. Workmanship not as specified.

Minor

201. Identification marking missing, incomplete, or illegible.
202. Instruction plates not as specified.
203. Treatment and painting not as specified.

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4.5.2 Tests.

4.5.2.1 Test conditions. Prior to test, the pump units shall be lubricated with oils and greases specified herein. Oils shall be those designated for use in the ambient temperature at the place of test. Tests shall be performed in accordance with the applicable provisions of the test code for reciprocating pumps of the Rating Standards and Test Code Standards of the Hydraulic Institute. Water shall be used as the test liquid, and all test data shall be corrected to standard sea level atmospheric conditions of 29.92 inches of mercury and to the specific weight of water (62.3 pounds per cubic foot) at a temperature of 68° F.

TABLE I. Test schedule

First article	Quality conformance		Test	Test paragraph	Requirement paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	-	Test conditions	4.5.2.1	-
X	-	-	Safety	4.5.2.3	3.5
X	-	-	Noise level measurement	4.5.2.4	3.6
			Reliability:		
X	-	-	Pump	4.5.2.5.1	3.7.1
X	-	-	Diaphragm	4.5.2.5.2	3.7.2
X	-	-	Maintainability	4.5.2.6	3.8
X	X	-	Priming	4.5.2.7	3.11
X	X	-	Hydrostatic pressure	4.5.2.8	3.11
			Pump unit performance:		
X	-	X	Size A pump unit	4.5.2.9.1	3.13
X	-	X	Size B pump unit	4.5.2.9.2	3.14
X	-	-	Engine performance (class 1)	4.5.2.10	3.15.1
X	-	-	Electromagnetic interference	4.5.2.11	3.16
X	-	-	Slinging provisions	4.5.2.12	3.18
X	-	-	Tiedown provisions	4.5.2.13	3.19

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4.5.2.3 Safety. The pump units shall be evaluated for safety requirements throughout testing as specified in 4.5.2. Nonconformance to 3.5 shall constitute failure of this test.

4.5.2.4 Noise level measurement.

4.5.2.4.1 Representative duty position (class 1). Measure the noise level at every 45 degrees plus or minus 5 degrees of azimuth, 65 inches plus or minus 1 inch above ground level, at a 15-foot radius from the geometric center of the equipment in accordance with MIL-STD-1474 for steady-state noise. The operating conditions shall be as specified in 4.5.2.9 and the test surface shall be compact earth and gravel or paved. Nonconformance to 3.6 shall constitute failure of this test.

4.5.2.4.2 Exterior (class 1). Measure the noise level in accordance with MIL-STD-1474 for exterior noise. Nonconformance to 3.6 shall constitute failure of this test.

4.5.2.4.3 Hazard precautions (class 2). Measure the noise level at all control and maintenance positions while operating at the maximum noise condition for these locations. Should the noise level exceed 85 dB(A) at any position, failure to provide the precautions specified in 3.6 shall constitute failure of this test.

4.5.2.5 Reliability.

4.5.2.5.1 Pump. Using the MTBF specified in 3.7.1, the pump unit shall be tested as specified herein, with "accept" and "reject" criteria in accordance with figure 1. Continue testing until either an "accept" or "reject" decision is reached. A failure is defined as any malfunction which cannot be corrected within 30 minutes (all waiting periods are excluded) by adjustment, repair or replacement action using controls and on-equipment tools or parts available to the using element and is a malfunction which causes or may cause one or more of the following:

a. Failure to commence operation, cessation of operation, or degradation of performance below the designated levels.
b. Serious damage to the system.
c. Serious personnel safety hazards.

4.5.2.5.1.1 Test procedure. With a static suction lift equivalent to 10 feet and using a 3-inch suction hose for the size A pump and a 4-inch suction hose for the size B pump, the pump unit shall be operated at 60-strokes-per-minute open discharge continuously for a period of 23 hours out of every 24 hours. Normal maintenance shall be conducted during scheduled downtime. Operations and maintenance shall be performed in accordance with the instructions set forth in the contractor's operation and maintenance schedule.

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4.5.2.5.1.2 Failure criteria. Nonconformance to 3.7.1 shall constitute failure of this test.

4.5.2.5.2 Diaphragm. During testing of the pump units as specified in 4.5.2.5.1, the number of diaphragm failures, if any, shall be recorded. Total test time shall be expressed in multiples of specified MTBF (see 3.7.2) and the test shall continue until such time as an "accept" or "reject" decision is reached. Nonconformance to 3.7.2 shall constitute failure of this test.

4.5.2.6 Maintainability.

4.5.2.6.1 Maintenance support evaluation. Maintenance operations required during reliability testing shall be accomplished to determine conformance to 3.8.1. Failure of the support elements to maintain the pump unit as specified shall constitute failure of this test. Errors or inadequacies in the manuals shall not be considered in assessing the maintenance ratio or the MTBF.

4.5.2.6.2 Maintenance ratio. The maintenance ratio shall be computed during reliability testing. Nonconformance to 3.8.2 shall constitute failure of this test.

4.5.2.7 Priming. The priming test shall be conducted with the center of the pump inlet 25 feet above the level of the water. A 30-foot suction line of the same nominal size shall be connected to the pump inlet and the discharge shall be open to the atmosphere. It is permissible to first fill the pump body with water. The pump shall be operated at not more than 60 strokes per minute. Failure of the pump to prime itself within 5 minutes shall constitute failure of the test.

4.5.2.8 Hydrostatic pressure. The entire pump, including the suction passageway, suction-valve chamber, diaphragm chamber, discharge-valve chamber, and discharge passageway, shall be subjected to a hydrostatic pressure of 50 psi for 2 minutes. During the hydrostatic test, the diaphragm shall be in the position that gives maximum diaphragm chamber volume. Water leakage through any part of the pump shall constitute failure of the test.

4.5.2.9 Pump unit performance.

4.5.2.9.1 Size A pump unit. The pump unit shall be operated at 60 strokes per minute with static suction lifts equivalent to both 10 and 20 feet at standard conditions. Failure of the pump to deliver 50 gpm at the 10-foot suction lift conditions and 33 gpm at the 20-foot suction lift conditions shall constitute failure of this test.

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4.5.2.9.2 Size B pump unit. The pump unit shall be operated at 60 strokes per minute with static suction lifts equivalent to both 10 and 20 feet at standard conditions. Failure of the pump to deliver 100 gpm at the 10-foot suction lift conditions and 66 gpm at the 20-foot suction lift conditions shall constitute failure of this test.

4.5.2.10 Engine performance (class 1). Prior to examination and tests of the first article pump unit, the engine that will be installed in the first article model unit shall be tested and sealed in accordance with MIL-STD-1400, test method series 2000, except that environmental test method 2400 will not be required. During the performance test specified in 4.5.2.9 the engine speed (rpm) and intake manifold vacuum shall be measured and recorded at the beginning and end of each performance run. These readings shall be compared with the engine speed (rpm), brake horsepower, and intake manifold vacuum curve for the condition of MIL-STD-1400, test method 2300, at the corresponding speed (rpm), shall constitute failure of this test.

4.5.2.11 Electromagnetic interference. The class 1 first article model shall be tested to determine conformance with 3.16. The contractor shall furnish the contracting officer a report of the test required by MIL-STD-461. Disapproval of the report shall constitute failure of this test.

4.5.2.12 Slinging provisions.

4.5.2.12.1 Test procedure.

a. Lift the pump unit and hold it in suspension in its normal travel position, using slings that converge not more than 24 feet above the lowest extremity of the pump unit. Determine the forward angles of application for each provision. Measure the clearance between each sling and the pump unit. Measure the eye openings and clearance dimensions of the provisions.

b. Restrain the pump unit by anchoring the main frame and subject each slinging provision to a force of 2.5 times the load which the unit carried when suspended. This force shall be applied in the direction as determined in "a" above. An acceptable alternate method of test shall be to lift the pump unit as in "a" above and add weights to the pump unit until the force on each provision is 2.5 times the initial force. Hold the pump unit under either of the above loading conditions for not less than 90 seconds.

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4.5.2.12.2 Failure criteria. Weld failure, permanent deformation of the pump unit or any slinging provision, slings converging more than 24 feet above the lowest extremity of the pump unit, or nonconformance to MIL-STD-209 shall constitute failure of this test.

4.5.2.13 Tiedown provisions. With the pump unit anchored by means other than the tiedown being tested, apply the static loads specified in MIL-STD-209 to each provision. Measure the dimensions of the tiedowns. Weld failure, permanent deformation of the pump unit or its tiedown provisions, breakage of provisions, or nonconformance to MIL-STD-209 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 First article pack inspection. The first article pack shall be examined for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

No.	Defect	A	B	Comm
114.	Disassembly not as specified.	5.2.1.1	5.2.2	
115.	Matchmarking not as specified.	5.2.1.?	5.2.2	
116.	Exterior surfaces not preserved as specified.	5.2.1.4	5.2.2	5.2.3
117.	Engines not preserved as specified.	5.2.1.5	5.2.2	5.2.3.1
118.	Pumps not preserved as specified.	5.2.1.6	5.2.2	5.2.3
119.	Data items not preserved as specified.	5.2.1.7	5.2.2	
120.	Repair parts not preserved as specified.	5.2.1.8.1	5.2.2	5.2.3
121.	Maintenance tools not preserved as specified.	5.2.1.8.2	5.2.2	5.2.3
122.	Hoses (when required) not preserved as specified.	5.2.1.9	5.2.2	5.2.3
123.	Consolidation container not as specified.	5.2.1.11	5.2.2.1	
124.	Components within consolidation containers not blocked and cushioned as specified.	5.2.1.11	5.2.2.1	

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No.	Defect	A	B	Comm
125.	Tool box (as applicable) not utilized as specified.	5.2.1.11	5.2.2.1	
126.	Shipping container not as specified.	5.3.1	5.3.2	5.3.3
127.	Consolidation container not in shipping container as specified.	5.3.1	5.3.2	
128.	Shipping container contents not blocked, braced, and anchored in position.	5.3.1	5.3.2	5.3.3
129.	Strapping not as specified.	5.3.1	5.3.2	
130.	Marking not as specified.	5.4.1	5.4.1	5.4.2
131.	Depreservation guide not attached to the unit as specified.	5.5	5.5	5.5

5. PACKAGING

5.1 First article pack. Unless otherwise specified (see 6.2), the contractor shall furnish a preproduction pack for examination within the time frame specified (see 6.2) to prove prior to the start of production packaging that the applied preservation, packing, and marking comply with the requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.11). The preproduction pack may be prepared by utilizing either the first article model or a production pump unit. When the first article model is utilized and the Government requests a comparison between the first article model and a production pump unit, any preservation and packing shall be removed by the contractor at no expense to the Government.

5.2 Preservation. Preservation shall be level A, level B, or commercial, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Disassembly. Disassembly of the pump units shall be the minimum necessary to safeguard parts vulnerable to damage, pilferage, and loss. The extent of further disassembly shall be confined to the removal of protruding parts that would otherwise increase cubage, however, disassembly shall be limited to those parts that can be removed and reinstalled without special tools or skilled personnel. Removed bolts, nuts, pins, screws, and washers shall be reinstalled to mating parts and secured to prevent their loss.

5.2.1.2 Matchmarking. Removed parts and mating parts on the pump units shall be matchmarked when necessary to facilitate reassembly. Matchmarking information shall be marked on cloth shipping tags conforming to UU-T-81, type A, grade optional and the marked tags shall be attached to mating parts with wire or twine.

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5.2.1.3 Preservative applicability. The provisions of MIL-P-116 shall apply when determining the applicability of preservatives. The preservatives specified herein by "P" numbers shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

5.2.1.4 Exterior surfaces. Unpainted exterior ferrous metal surfaces of bolts, nuts, pins, screws, washers, brackets, tanks, and any other exterior surface requiring a contact preservative (see 5.2.1.3) shall be coated with type P-1 preservative. Unpainted ferrous metal information plates shall be coated with varnish conforming to MIL-V-13811 or TT-V-119.

5.2.1.5 Engines. Engines and engine accessories shall be preserved in accordance with the level A requirements of MIL-E-10062, type II, method I.

5.2.1.6 Pumps. The interior surfaces of the pump casings and all parts within the casings shall be coated with preservative lubricating oil conforming to P-10, type I, grade 30. The suction and discharge openings shall be sealed, as applicable, with plastic or metal caps, plugs, or blank flanges. Alternatively, the openings shall be covered with barrier material conforming to MIL-B-121, type I, grade A, class 2 securely held in place with tape conforming to PPP-T-60, type IV.

5.2.1.7 Data items. Data items for each pump unit (see 6.9) shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

Special note: When more than one data item (such as parts list, operation manual, maintenance manual or others) is required for each pump unit, all such items shall be preserved as a set in one method IC-1 or IC-3 bag.

5.2.1.8 Provisioning items (see 6.10).

5.2.1.8.1 Repair parts. Repair parts for each pump unit shall be preserved in accordance with the level A requirements of MIL-R-196. Parts not specifically mentioned therein shall be preserved as specified for parts of similar design and construction.

5.2.1.8.2 Maintenance tools. Maintenance tools shall be preserved in accordance with the level A requirements of PPP-P-40.

5.2.1.9 Hoses. When required (see 6.2), hoses and hose couplings and fittings shall be preserved in accordance with the level A requirements of MIL-P-775.

5.2.1.10 Other components. Other components not specifically mentioned herein requiring protection from corrosion or physical or mechanical damage shall be preserved as specified for components of similar design and construction.

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5.2.1.11 Consolidation. Loose small components, provisioning items, data items, hoses (when required), and maintenance tools shall be consolidated into a close fitting box conforming to PPP-B-636, type CF, class weather-resistant. When tool boxes are provided, the tool box shall be utilized in so far as space permits for the consolidation of small items. The tool box lid shall be closed and secured. The remainder of the components that will not fit in the tool box shall be consolidated with the tool box into a snug fitting box conforming to PPP-B-636, type CF, class weather-resistant. The contents shall be blocked and cushioned to prevent or control movement within the container.

5.2.2 Level B. Level B preservation shall be as specified for level A (see 5.2.1), except for the following:

5.2.2.1 Consolidation. Consolidation shall be as specified in 5.2.1.11 for level A except that the box shall be class Domestic.

5.2.3 Commercial. Each complete pump unit with loose components, provisioning items, hoses (when required), and maintenance tools (except for engine and engine accessories (see 5.2.3.1)), shall be preserved in accordance with MIL-STD-1188.

5.2.3.1 Engine. All lubrication and fluid reservoirs shall be filled to the operating level with the lubricant or fluid designated for use in the temperature range at the shipping destination. A tag or tags, as applicable, shall be attached in a conspicuous location indicating which lubricant or fluid has been used.

5.3 Packing. Packing shall be level A, level B, or commercial as specified (see 6.2).

5.3.1 Level A. Each complete pump unit, preserved as specified in 5.2.1, shall be packed in a box conforming to PPP-B-601, overseas type, style A or B or PPP-B-621, class 2, style 7 with hood. The pump unit and its consolidation container shall be arranged within the box in a manner to assure minimum cube with blocking, bracing, and anchoring as specified in MIL-STD-1186. Box closure and strapping shall be specified in the applicable box specification and/or the appendix thereto. Strapping shall be flat and zinc coated.

5.3.2 Level B. Each complete pump unit, preserved as specified in 5.2.2, shall be packed as specified for level A (see 5.3.1) except that boxes shall be domestic type or class and the flat strapping need not be zinc coated.

5.3.3 Commercial. Each complete pump unit, preserved as specified in 5.2.3, shall be packed in accordance with MIL-STD-1188.

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5.4 Marking.

5.4.1 Levels A and B. In addition to any special or other identification marking required by the contract or order (see 6.2), consolidation containers and shipping containers for each pump unit packed as specified in 5.3.1 and 5.3.2, shall be marked in accordance with MIL-STD-129.

5.4.2 Commercial. Each pump unit packed as specified in 5.3.3 shall be marked in accordance with MIL-STD-1188.

5.5 Depreservation guide. A depreservation guide shall be prepared, placed in a waterproof envelope marked "Depreservation Guide" and secured in a conspicuous location on the pump. Unless otherwise specified (see 6.2 and 6.12), DA Form 2258 "Depreservation Guide for Vehicles and Equipment" shall be used.

6. NOTES

6.1 Intended use. The pump unit is intended for use in dewatering applications where the water bears sand, gravel, or any other solids.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Class and size of pump required (see 1.2).

c. When a first article is required for inspection and approval and the number of units required (see 3.2).

d. When a gear train is required to transmit power from the engine (see 3.10).

e. When wheels with cushion tires not less than 10 x 2.75 x 2 ply are required (see 3.12).

f. When hoses shall be furnished with the size B pumps (see 3.14.3).

g. When diaphragm-type fuel pump is required (see 3.15.1.2).

h. When electromagnetic interference emission of the class 1 pump units shall be other than as specified (see 3.16).

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i. When lubricants as recommended by the manufacturer are required (see 3.17.1).

j. When grease lubrication with other than the specified greases is required (see 3.17.4).

k. When oil lubrication other than as specified is required (see 3.17.5).

l. Welding process required if other than the manufacturer's standard process (see 3.24.3).

m. When a preproduction pack is not required (see 5.1).

n. Time frame required when a preproduction pack is required (see 5.1).

o. Degree of preservation and degree of packing required (see 5.2 and 5.3).

p. When hoses, hose couplings, and fittings shall be preserved in accordance with MIL-P-775 (see 5.2.1.9).

q. When additional marking is required (see 5.4.1).

r. When other than DA Form 2258 is required (see 5.5).

6.3 First article. When a first article inspection is required, the item shall be tested and should be a first produced pump unit. The first article should consist of two or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test, and approval of the first article.

6.4 Definition. Standard commercial product is a product which has been or will be sold on the commercial market through advertisements or manufacturer's catalogs or brochures, and represents the latest production model(s) (see 3.1).

6.5 Noise limits. When the noise level of the pump unit exceeds 85 dB(A), the appropriate discussion of the noise hazard in accordance with MIL-STD-1474 (Manuals Paragraph) shall include the requirement for hearing protection, the noise level of the pump unit, and the distance from the pump unit at which 85 dB(A) will always be met.

6.6 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the pump units covered by this specification or contained in the Federal Supply Catalog, Department of Defense Section, Identification List C9100-IL, for FSC Group 91.

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6.7 Government-furnished property. The contracting officer should arrange to furnish the property specified in 3.15.2.

6.8 Power requirements. To determine the capability of the engine to comply with the power and speed requirements of 3.15.2.3, the contracting officer may arrange to have any or all of the first article tests conducted on the dynamometer.

6.9 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each pump unit.

6.10 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of pump units.

6.11 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack, and mark the pump units in accordance with this specification (see 5.1).

6.12 DA Form 2258. The contracting officer should arrange to furnish DA Form 2258 when requested by the contractor (see 5.5).

6.13 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

Custodians:
Army - ME
Navy - YD
Air Force - 99

Review activity:
Navy - MC

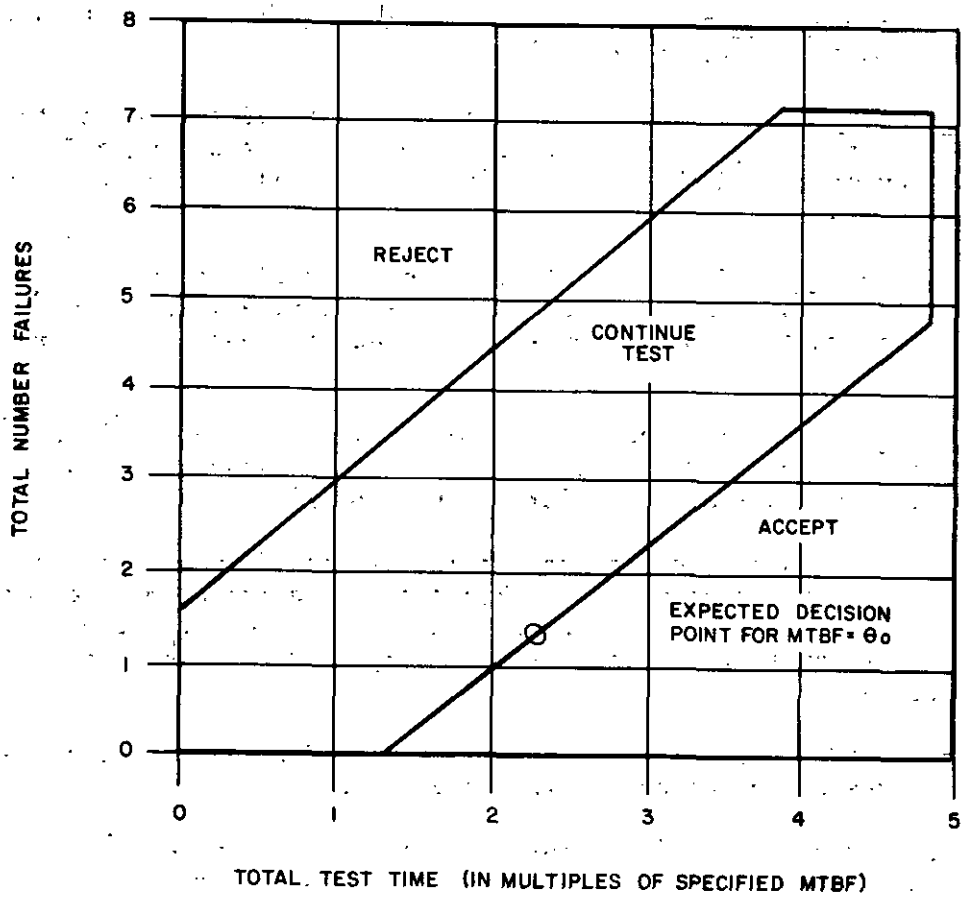
User activity:
Air Force - 84

Preparing activity:
Army - ME

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DECISION RISKS 20%
DISCRIMINATION RATIO 2.0:1



NO. OF FAILURES	TOTAL TEST TIME* REJECT (EQUAL OR LESS)	ACCEPT EQUAL OR MORE
0	N/A	1.40
1	N/A	2.09
2	.35	2.79
3	1.04	3.48
4	1.73	4.17
5	2.43	4.87
6	3.12	4.87
7	3.81	4.87
8	4.87	N/A

*TOTAL TEST TIME IS TOTAL UNIT HOURS OF "EQUIPMENT ON" TIME AND IS EXPRESSED IN MULTIPLES OF THE SPECIFIED MTBF.

FIGURE 1. Accept-reject criteria.

X-3661

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-509F	Pumps, Reciprocating, Power-Driven: Diaphragm, Gasoline-Engine-Driven, Wheel-Mounted

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

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